

Finite State Machines

Tutorial 6 - Digital Design (CS F215)

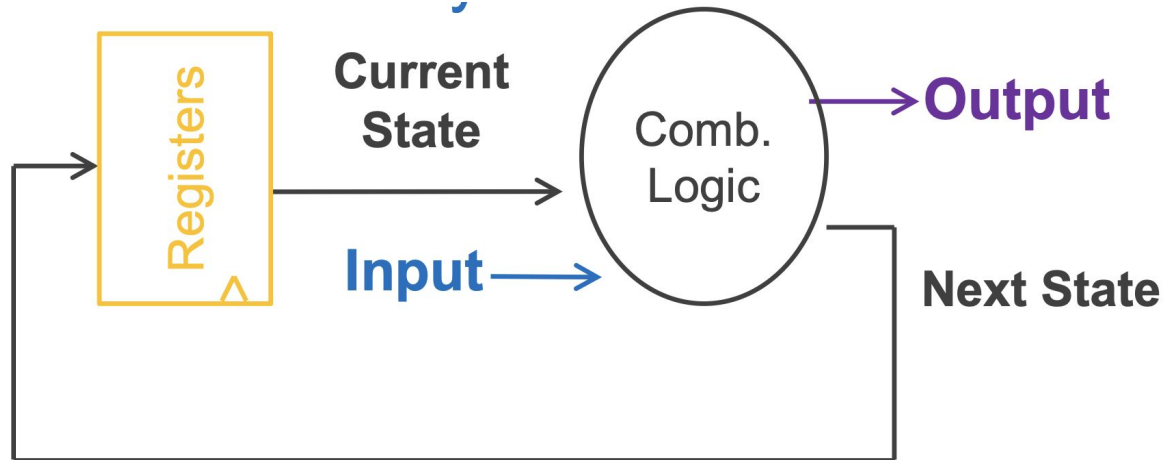
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Why FSM?

- Many real-world systems change behavior based on current situations (states and inputs).
- Traffic lights, vending machines, elevators, follow a sequence of states.
- FSM helps us model such systems clearly and logically.
- It captures “what to do now” and “when to switch”.

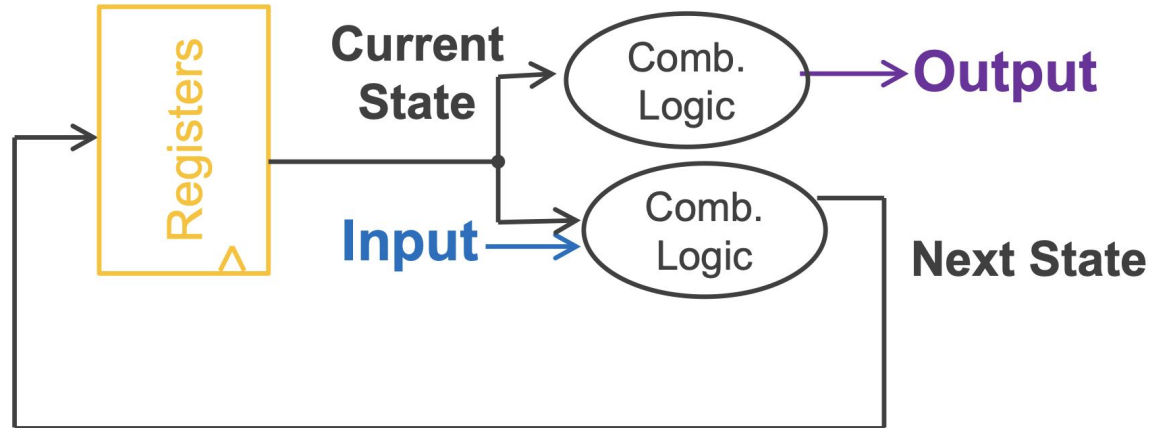
Types of FSM - Mealy Machine

- Output depends on both state and input - reacts instantly to changes.
- Useful when immediate response is needed (e.g., input-driven control).
- Example: An automatic door that opens instantly when a person is detected, i.e., output changes immediately with input.

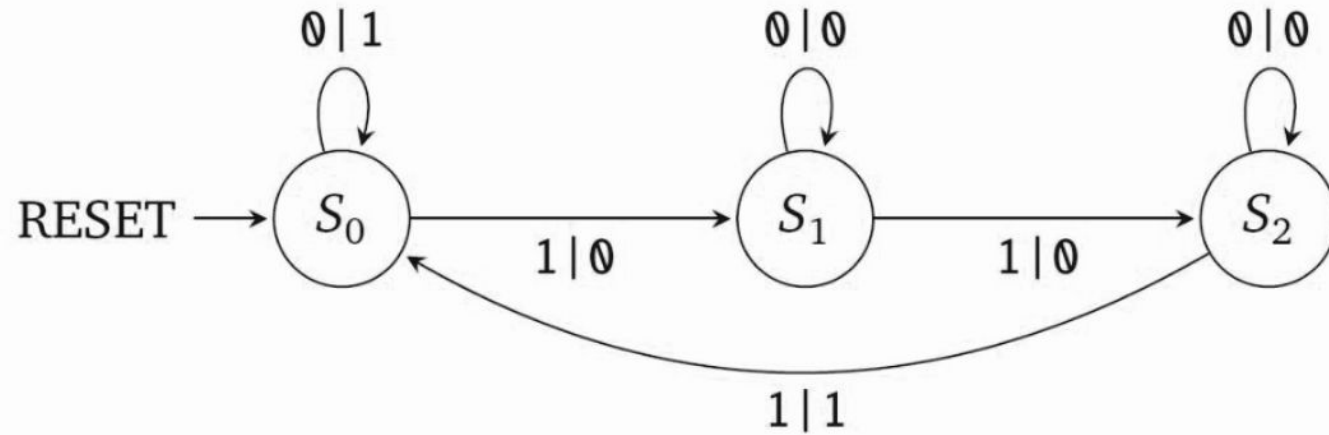


Types of FSM - Moore Machine

- Output depends only on the current state, not immediate inputs.
- Easier to design and debug since outputs change only on state transitions.
- Example: A traffic light controller, output (light color) depends on the current state, not sensor input.



Code Along



Thank you!

Questions?